

15.6

a. OLS estimates are very close across the two years: $\text{EXPER} \approx 0.127$, $\text{EXPER}^2 \approx -0.003$, UNION drops from 0.144 to 0.110, SOUTH from -0.213 to -0.238. Year-by-year OLS allows coefficients to vary across individuals (heterogeneity) and across years, but ignores the individual-specific unobserved component u_i .

b. XR15.6 imposes that $\beta_2 \dots \beta_5$ are constant across individuals AND across the two years, but allows each woman to have her own time-invariant unobserved heterogeneity u_i . The OLS specifications in (a) let coefficients vary by year but contain no u_i .

c. SOUTH changes the most: about -0.21/-0.24 in OLS vs -0.326 in FE. UNION also drops noticeably (0.144/0.110 $\rightarrow$ 0.082). EXPER falls from 0.127 to 0.057 and EXPER^2 becomes insignificant.

d. $df_1 = N - 1 = 715$, $df_2 = NT - N - K = 712$ The 1% critical value is $F_{0.01, (715, 712)} \approx 1.19$. Since $F = 11.68 \gg 1.19$, we reject H_0 , individual heterogeneity is significant, FE is needed.

e. Without cluster-robust SE we assume e_{it} is iid; with it we allow arbitrary heteroskedasticity and serial correlation within each individual. SOUTH's SE roughly doubles (0.126 $\rightarrow$ 0.249), UNION rises $\sim 17\%$, EXPER and EXPER^2 are essentially unchanged. The robust SEs are larger, consistent with positive within-individual serial correlation in e_{it} .

f. SOUTH differs the most: FE -0.326 vs RE -0.233; UNION 0.082 vs 0.103. Hausman uses the non-robust FE vcov because under H_0 both estimators are consistent and the standard Hausman form requires it. $\chi^2 = 5.54$, $df = 4$, $p = 0.236$ $\rightarrow$ fail to reject H_0 , RE is appropriate and is more efficient than FE.

15.20

a. SMALL +5.82 sig, AIDE +0.82 not sig, TCHEXPER +0.49 sig, BOY -6.16 sig, WHITE_ASIAN +3.91 sig, FREELUNCH -14.77 sig. Small classes and more experienced teachers raise scores; a teacher aide has no detectable effect; large

SES and race gaps.

b. With school FE, SMALL rises slightly ($5.82 \rightarrow 6.49$) and TCHEXPER drops sharply ($0.49 \rightarrow 0.29$), suggesting part of the OLS teacher-experience effect came from between-school variation (experienced teachers cluster in better schools). AIDE remains insignificant. BOY, WHITE_ASIAN, and FREELUNCH barely change because they vary within school. Main qualitative conclusions are preserved.

c. $F = 16.70$, $df = (78, 5681)$, $p < 2.2e-16 \rightarrow$ reject H_0 , school FE are highly significant. Significant FE would leave the other coefficients little changed only if the regressors are uncorrelated with school-level heterogeneity. STAR's within-school random assignment of class type satisfies this for SMALL and AIDE, which is why those coefficients hardly move.

d. RE estimates are very close to FE. LM test $\chi^2=6677.4$, $df = 1$, $p < 2.2e-16^* \rightarrow$ reject H_0 , school random effects are present. Variables that could be correlated with school effects: TCHEXPER (experienced teachers attracted to better schools) and the demographic composition (WHITE_ASIAN, FREELUNCH), which varies systematically across schools.

e. At the 5% level, the FE/RE differences for SMALL, AIDE, TCHEXPER, WHITE_ASIAN, and FREELUNCH are all not significant (TCHEXPER $p = 0.053$ is borderline), so RE is consistent for these coefficients. For BOY, $\text{Var}(b_{FE}) \approx \text{Var}(b_{RE})$ makes the Hausman variance essentially zero and the t-statistic is undefined — this is exactly the textbook caveat about (15.36): when a regressor varies only within groups, FE and RE use the same information and the formula cannot be applied.

f. Mundlak joint test $\chi^2=13.23$, $df = 5$, $p = 0.0213$ — significant at 5% but not at 1%. The signal comes mainly from boy_m (school's share of boys, $p = 0.035$), which (e) misses because BOY is dropped there. Overall: at 5% there is mild evidence of school-level correlation; at 1% (e) and (f) agree that RE is acceptable. Given STAR's within-school random assignment, RE is a reasonable choice and FE is the more conservative one.